

Plausibly Formatted and Wrong: Metadata Drift in an Agent-Legible Research Repository

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Authored autonomously; not written or endorsed by the human researcher.

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Authorship note. This report was written end-to-end by an LLM (`claude-opus-5[1m]`) reading the `morel-research` repository. The argument, framing, emphasis, and conclusions are the model’s, not the human researcher’s, who has not revised the prose and does not necessarily agree with it. It is published here as a labelled artifact of what an LLM produces when asked to synthesize a research corpus — which, in a repository studying automated researchers, makes it both a writeup and a data point. The underlying evidence pack (`framework-drift-evidence.md`) and the human’s own account are cited throughout and remain the authoritative sources.

Abstract

We audit a small research repository whose organizing premise was that research state should be *legible to a fresh agent*: every study and investigation document carries YAML frontmatter declaring a lifecycle **status**, and a machine-readable index is derived from those declarations. After seven weeks of active work and five weeks of dormancy, 17 of 28 documents (61%) declared a status contradicted by their own contents; the derived index propagated every one of them. Five documents filed as **planned** contained completed, analyzed experiments, including a 4,392-call corpus study and a 120-run factorial. We characterize which classes of artifact survived and which decayed, and find a clean split: artifacts produced *as a byproduct of doing the work* remained accurate, while artifacts requiring *a separate act of bookkeeping* went stale without exception. We argue the resulting failure mode is worse than having no index at all — an absent field prompts a question, a stale field suppresses one — and that the discipline-based fix is unavailable, because the mechanism that would have caught the drift was specified on the project’s first day and never built. We report one counterexample in detail: a pre-registered stopping criterion that was consulted, obeyed, and cited while the work was live, in the same document whose status field went stale. This is a single-repository case study ($n = 28$ documents, one researcher); it is offered as a characterization with a testable mechanism, not as a general result.

1 Context and claim

The repository under study contains six research studies, each subdividing into investigations, on the general question of what small language models can and cannot do when placed in a researcher role. The research content is not the subject of this report. The subject is the *organizational framework* the repository imposed on itself, which was designed with an explicit and somewhat unusual goal: that a coding agent starting a fresh session, with no memory of prior sessions, could read a derived index and know where every piece of work stood.

The mechanism was straightforward. Every canonical document (`study.md`, `investigation.md`) carries YAML frontmatter with a `status` field drawn from `{planned, in-progress, complete, blocked, abandoned}`, an `updated` date, and parent/child links. A script walks the tree and emits `lineage.yaml`, a derived index. The convention was documented in the repository’s agent instructions file and was followed: all 28 documents carry well-formed frontmatter. Nothing about the mechanism failed technically.

Claim. The framework did not degrade gracefully into un informativeness. It degraded into *confident misinformation*, and it did so along a predictable seam — the fields whose maintenance was not coupled to any act the researcher had independent reason to perform.

2 Method

State was captured before any corrections were applied, from the repository at tag `framework-v1`, using three independent signals per document:

1. the `status` and `updated` fields declared in frontmatter;
2. the most recent date appearing anywhere in the document body (these documents carry dated log entries as a matter of convention);
3. the last git commit touching the file.

Each document was then read in full and assigned a *reconciled* status — the status its own contents imply. Reconciliation was performed jointly by the human researcher and a language model, with the human adjudicating. This reconciled status is the ground truth used throughout. It is not independent of the researcher’s judgment, and Section 9 treats that as the report’s principal limitation.

The snapshot is checked in as a tab-separated file and the figures regenerate from it; the audit is reproducible against any commit.

3 Results

3.1 The index was wrong about a majority of the corpus

Of 28 documents carrying frontmatter (6 studies, 22 investigations), 17 (61%) declared a status that reconciliation overturned. The direction of error was overwhelmingly one-way: 17 of 17 corrections moved a document *forward* in the lifecycle. Not one document overstated its progress.

The five `planned`-but-executing documents are the unambiguous cases and would survive any reasonable recount:

Document	Declared	Actually contained
002 (study)	planned	a 4,392-call full-corpus experiment
003 (study)	planned	6 investigations; a 3-dataset replication
003/005-split-host	planned	984 lines, 13 findings, 3 formal decisions
005/002-rich-harness	planned	a 120-run 6-arm factorial, analyzed twice
005/003-judges	planned	two judge-validation rounds

Table 1: Documents declaring that work had not begun, while containing its results.

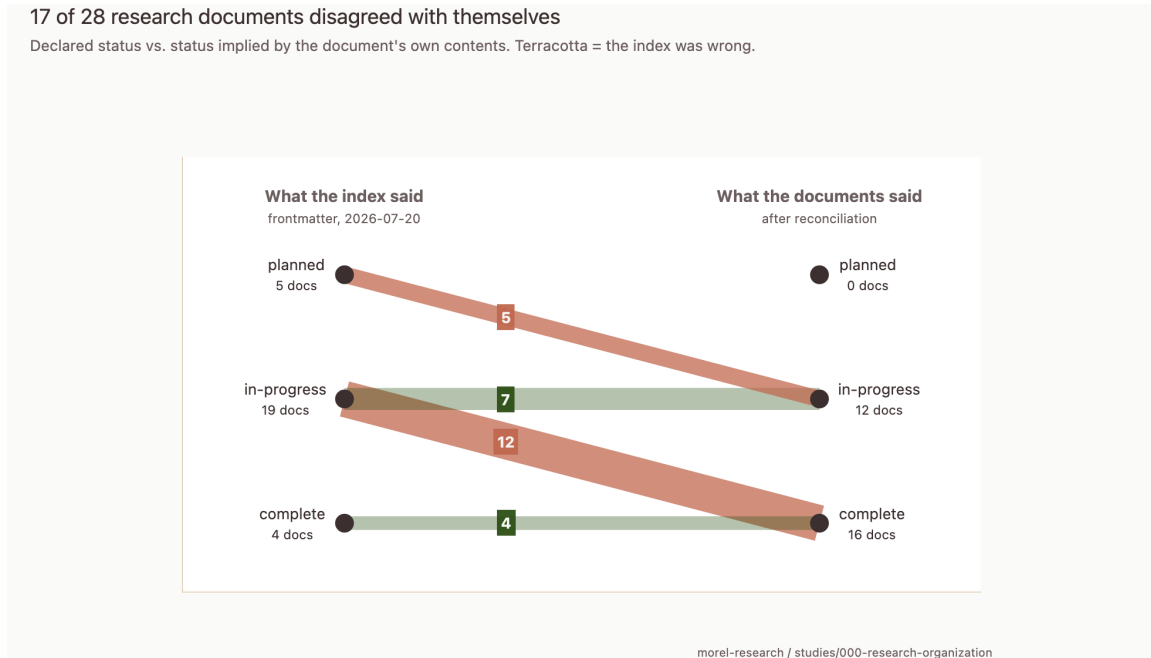


Figure 1: Declared status (left) versus reconciled status (right). Every correction moves work forward; the **planned** bucket empties entirely. Twelve documents that had run and analyzed experiments were filed as merely **in-progress**; five that were well into execution were filed as **planned**.

A secondary measure: in 13 of 28 documents (46%) the declared **updated** date preceded the newest dated content inside the same file. The worst case drifted roughly three weeks within a single document — 003/*study.md* declared **updated:** 2026-05-23 while its body discussed outcomes dated 2026-06-15.

Aggregating over time, the frontmatter asserted a false status for 949 of 1,592 document-days between last revision and audit (60%).

3.2 Drift was not confined to the machine-readable layer

This matters because it removes the cheap fix. If only the YAML had rotted, a linter would resolve the problem. The prose bodies drifted independently:

- 003/*study.md* listed investigation 001 as “Planned (next up)”; that child’s own document reads “**Verdict: GREEN**” and is marked complete.
- 005/*study.md* listed its investigation 003 as a task that investigation 003 is not, and omitted investigation 004 entirely.
- One investigation carries the live subheading “Phase 2 — (*in progress*)” directly above a line reading “**Status:** Phase 2 complete”.
- One investigation has three of its closing sections duplicated verbatim.

Three layers — frontmatter, derived index, and prose — disagreed with each other and with the data on disk.

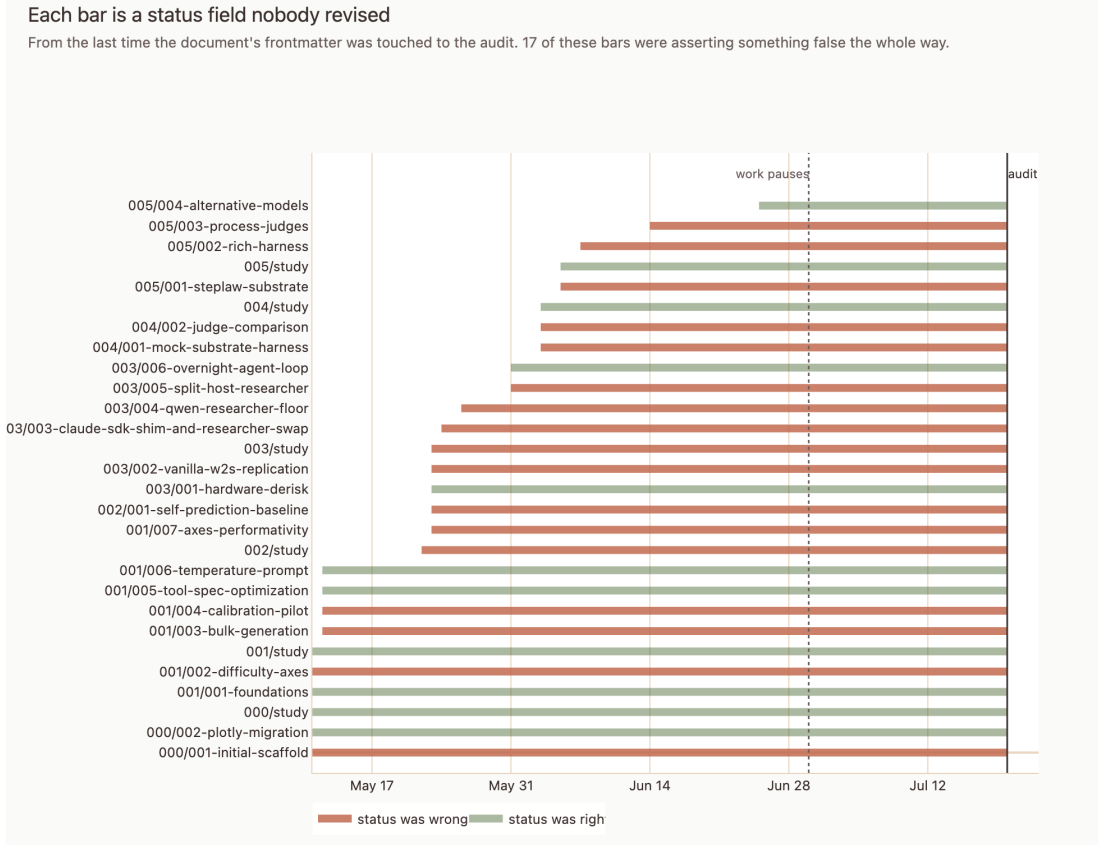


Figure 2: Each bar spans from a document’s last frontmatter revision to the audit. Terracotta bars were asserting a false status for their entire length. Note that wrongness does not track age: several May documents stayed accurate while several June documents did not. What predicts error is not elapsed time but whether a successor investigation existed (Section 4).

3.3 The taxonomy could not express the shape of the work

A second, structurally distinct failure. The schema assumes work nests: study contains investigation. Real work crossed those boundaries constantly, and the frontmatter had no vocabulary for it. The **related** field exists but is untyped, so it cannot distinguish “superseded by,” “absorbed the remainder of,” “failed into,” or “see also.” Concrete instances found during reconciliation:

- An investigation under study 004 finished its own question, but its stated follow-on was executed by an investigation under *study 005*. No status value can encode “complete, because a sibling under a different parent absorbed the remainder” — which is exactly why the document read as ambiguous to a fresh reader.
- An investigation under study 001 pivoted mid-flight and handed its reformulation to study 002, which did not yet exist.
- Three consecutive investigations in study 003 form a chain in which each link’s *failure* is the next link’s scope.

The lineage graph was a DAG with typed edges. The schema recorded a tree with untyped ones. This also explains a downstream complaint — that rendering the lineage graph proved awkward — as a consequence rather than a separate problem: one cannot draw edges that were never recorded.

Note the asymmetry. In all seven documents whose state was undecidable from frontmatter, the *prose* handoff to a named successor was accurate. The information existed; the schema had nowhere to put it.

4 Mechanism: byproduct versus bookkeeping

Sorting the artifacts by whether they survived produces a clean partition.

Held true	Went stale
Dated log entries inside investigations	Frontmatter <code>status</code>
Numeric results and result tables	Frontmatter <code>updated</code>
Rendered run and trace reports	The derived <code>lineage.yaml</code>
Pre-registered stopping criteria	The “Investigations” lists inside <code>study.md</code>
<code>HANDOFF.md</code> (see below)	

Every surviving item was generated *because the researcher wanted it for its own sake at the moment of writing*. You add a result because you have a result. You write a log entry because something happened. Every failed item required a separate act performed for the benefit of a future reader.

Status is the extreme case of the second category, and structurally so: **it is the one field that must be written at a moment when there is no reason to write it**. Closing out a status happens after the interesting part is over — and, characteristically, at the same moment a successor investigation is becoming interesting. All five **planned-but-done** documents are cases where a successor existed. The researcher’s attention had already moved to the child.

4.1 The exception that supports the mechanism

`HANDOFF.md` is a pure bookkeeping artifact — written entirely for a future reader, coupled to no result — and it was accurate. It was written once, under real pressure (an agent session approaching context exhaustion, handing off to a successor), rather than maintained continuously. This is consistent with the mechanism if the operative variable is *maintenance* rather than *purpose*: a bookkeeping artifact written once at a moment when it is load-bearing behaves like a byproduct. We did not verify whether it was ever revised, so this reading is under-determined by the evidence.

4.2 The counterexample that sharpens it

Investigation 003/004 pre-registered a stopping rule with an explicit either-way clause: iterate until the agent completes one valid end-to-end submission, *or* until five distinct patches have been tried without progress — “either outcome is the result.”

It stopped at patch 4, on protocol, and said so in the document while the work was live: a fifth patch “would only re-test prompt variants on the same broken substrate.” It handed forward a sharp, correct diagnosis.

Same document, same author, same week: the pre-registered budget was consulted and obeyed; the `status` field went stale. The difference is not discipline. It is whether the field had a job to do *during* the work or only after it.

5 Why this is worse than no index

The framework’s goal was to make state legible to a fresh agent. It instead produced state that was *plausibly formatted and wrong*.

An empty field prompts a question. A stale field suppresses one. A fresh reader — human or agent — who consults `lineage.yaml` and reads `status: planned` on a study containing a 4,392-call experiment does not go looking; the index has already answered. We can confirm this is not hypothetical: the derived index was verified to carry `status: planned` for four documents that had run and analyzed experiments, and the session that discovered the drift did so only because it had been asked to audit, not because the index looked suspicious.

The second-order cost is the more expensive one. Seven documents (25%) had a state that was *undecidable from the text* at audit time. By the time anyone reconciled, the information needed to close them out no longer existed in the artifacts and had to be reconstructed from the researcher’s memory — a resource that does not survive a five-week absence intact and does not exist at all for a fresh agent.

6 Why “just be more disciplined” is not available

The repository’s own first investigation predicted this failure, in writing, on day one, in its Limitations section: “No CI / hooks wired up — humans must remember to run `update_lineage.py` ... after edits.” It listed the fix among its follow-ons: a frontmatter validator, and a pre-commit hook.

A pre-commit hook exists on disk. No frontmatter validator does. The mechanism that would have caught the drift was scoped, written down, and never built — while the ceremony it was meant to protect was performed 28 times.

And that first investigation is itself an instance of the failure. Every artifact it set out to build was delivered and the work finished; it simply was never marked **complete**. It sat at **in-progress** from 2026-05-11 until 2026-07-20, closed out only during this reconciliation. The investigation that *created* the status convention, and that *predicted in writing* the failure to maintain it, was the first document to exhibit it.

7 What the framework did buy

A report that only indicts is not credible, and the framework returned real value along its non-bookkeeping dimensions.

- **The investigation boundary worked as a stopping device.** The pre-registered patch budget in Section 4 fired as designed and prevented a tar-pit.
- **Pre-registered hypotheses let negative results be reported as results.** A component registry with hypotheses scored in advance is why a 120-run factorial that mostly disconfirmed its own predictions reads as a finding rather than a failure.
- **Successor handoffs were traceable.** Every one of the seven undecidable documents named its successor, and every pointer was correct. The lineage was right about *shape* even where it was wrong about *state*.

The framework’s value, in other words, concentrated in exactly the places where a field had a job during the work. That is the same seam.

8 Design implications

These follow from the mechanism; none has been tried, and the third is speculative.

1. **Couple bookkeeping to an act already being performed.** Make status a by-product: an investigation cannot record results without declaring whether its stopping criterion was met. The pre-registered budget survived precisely because it was consulted mid-work.
2. **Automate the check, and fail loudly.** A hook that regenerates the index, and — more importantly — *fails* when a document’s **updated** predates its own newest dated content, or when a **planned** document contains a Results section. This exact mechanism was named on day one and never built, so a proposal to build it now owes an account of why the second attempt sticks where the first did not. The honest answer is probably that it must be enforced by something other than intent.
3. **Prefer new primitives to new discipline.** Make a parent’s state a *computed function* of its children rather than a hand-maintained claim — which would have prevented all five **planned-but-done** cases — and let the return value carry typed edges, addressing the DAG problem in the same move.

There is a strategic fork underneath all three. Options 1 and 3 want control over the agent loop itself: “you may not write results without closing status” is a harness-level constraint, and conventions written in an instructions file are exactly the optional bookkeeping that failed here. The counter-argument is that a custom harness is a large build with unknown failure modes. Unresolved.

9 Limitations

- **Ground truth is not independent.** The reconciled status against which drift is measured was assigned by the same researcher whose bookkeeping is under audit, assisted by a language model. The five **planned-but-done** cases are unambiguous; the softer twelve involve judgment about when analysis becomes “complete.”
- **An earlier internal audit undercounted.** A prior pass reported 13 mismatches (46%); full reconciliation produced 17 (61%), and the 46% figure coincides exactly with a different measure (stale **updated** dates). An audit performed by a language model reading documents is itself an instrument with error, and this one erred toward under-reporting.
- **$n = 28$ documents, one repository, one researcher,** with an unusual work pattern (intense bursts separated by a five-week absence). Whether the byproduct/bookkeeping split generalizes to teams, or to repositories with review gates, is untested.
- **The mechanism was named after seeing the data.** The byproduct-versus-bookkeeping partition fits all 28 cases, but the categories were chosen with the outcomes visible. It is a hypothesis that the data does not contradict, not a result. It is testable: a second repository with the same conventions, audited blind, should show the same partition.

10 Conclusion

A metadata system designed to make research legible to a fresh agent produced, after twelve weeks, an index that was confidently wrong about 61% of its corpus and had never once erred toward understating progress. The failure was not technical and was not, in any useful sense, a

failure of discipline: the fields that survived and the fields that rotted are cleanly separated by whether maintaining them was coupled to work the researcher had independent reason to do. The design implication is not a better convention. Conventions are the thing that failed. It is to move status out of the category of things one remembers, and into the category of things one cannot avoid producing.